Guidelines for hydraulic continuity verification

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1. Purpose & Scope

This procedure defines the criteria for determining whether it is necessary to check the geometric continuity of hydraulic passages, as well as the criteria for assessing measured deviations and deciding if corrective actions are required. It also outlines the procedures for carrying out the necessary activities when applicable.

The procedure is applicable to the plants of Nova Milanese and La Spezia and to pump types BB1 - BB3 - VS2 - VS7.

2. Definitions

BB1: Axially split, one- and two-stage, between-bearings pumps

BB3: Axially split, multistage, between-bearings pumps

ME: Manufacturing Engineering

Nq: Pump specific speed

PA: Pump Assembly

PT: Product Technology

TS: Technical Summary

VS2: Wet pit, vertically suspended, single-casing volute pumps

VS7: Double-casing, volute or volute plus diffuser, vertically suspended pumps

3. General Procedure - Description

In horizontally split case pumps (BB1/BB3), it is quite common to encounter misalignment of the hydraulic profiles of the volutes between the lower and upper casing halves at the joint plane (*Figure 1* and *Figure 2*), caused by inaccuracies in the casting process (unexpected core shifts, non-uniform shrinkage, etc.).

Such misalignments give rise to discontinuities within the profiles of the hydraulic channels, which may adversely affect pump performance. These discontinuities can have a significant impact on the measured efficiency.

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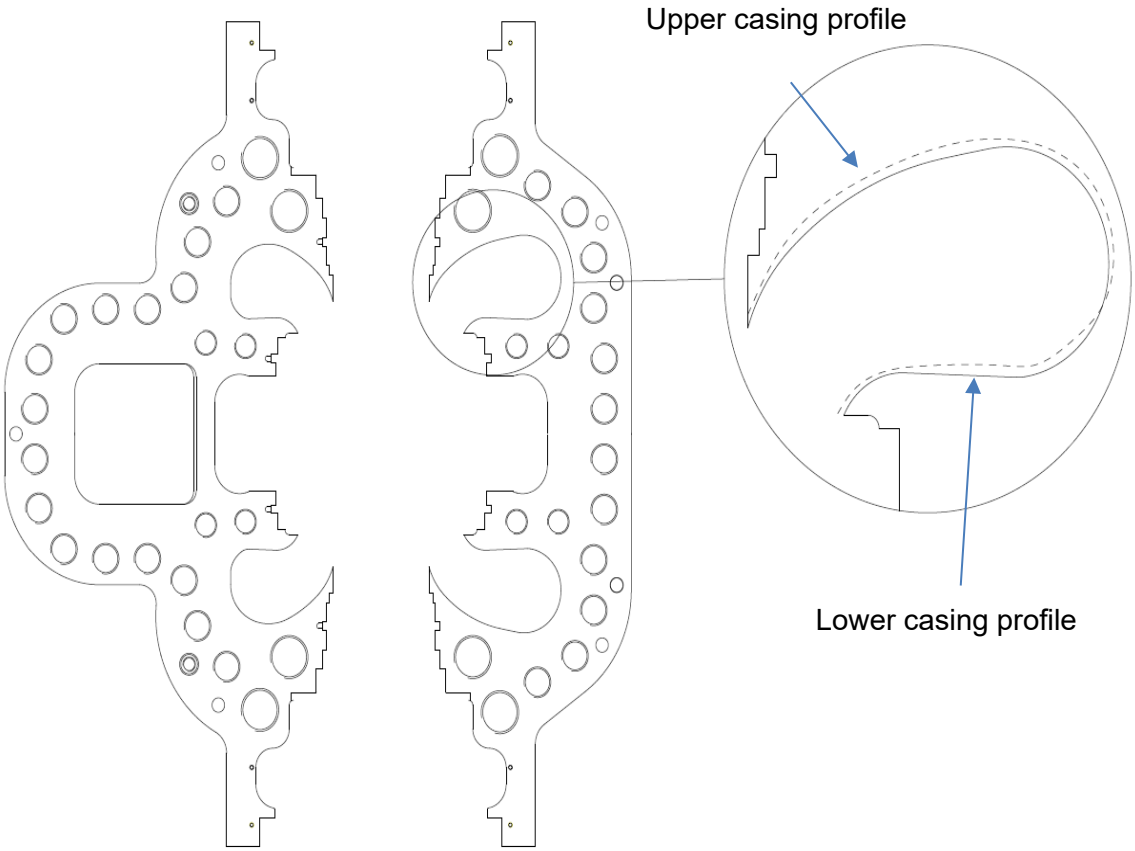

Figure 1:Example of hydraulic misalignment on BB3 pump casing.

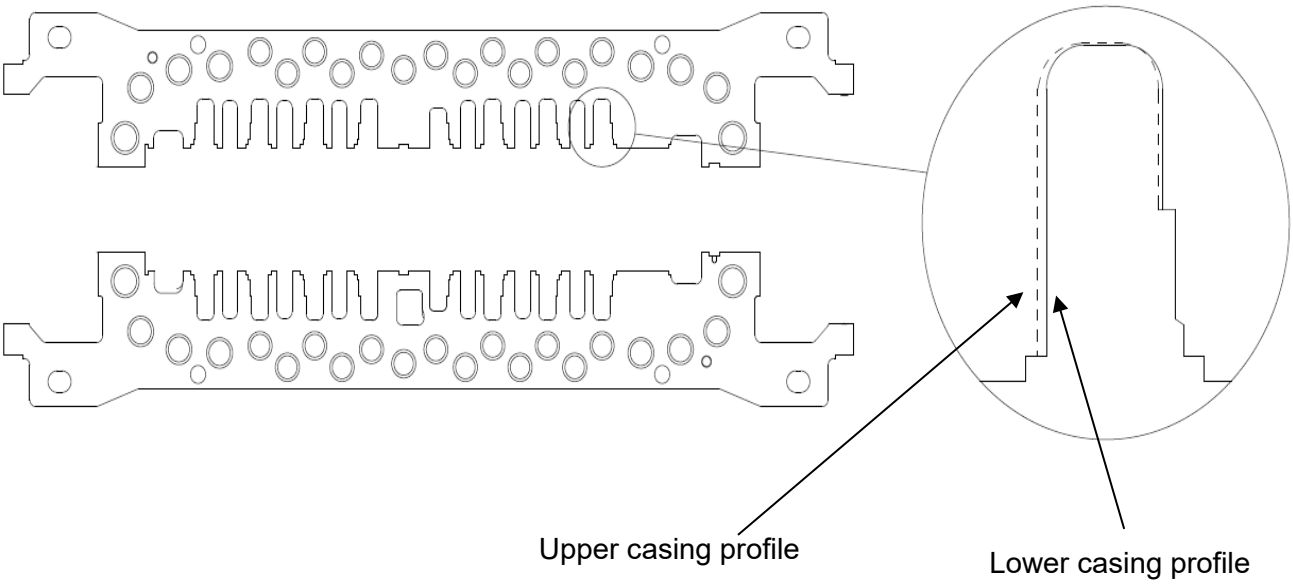

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Figure 2: Example of hydraulic misalignment on BB1 pump casing.

Similarly, for vertical pumps with a first-stage double suction impeller (VS2/VS7), when the discharge connection is a separate component, misalignments frequently occur in the hydraulic channel at the flanged connections between the first-stage volute and the discharge connection (*Figure 3*).

In this area, two independent flows must merge into a single stream and any discontinuity at the joint plane may result in velocity differences between the two branches, residual swirl, secondary flows, internal recirculation, asymmetric flow to the second-stage diffuser (if any), with efficiency losses in both the first and the second stage (if any).

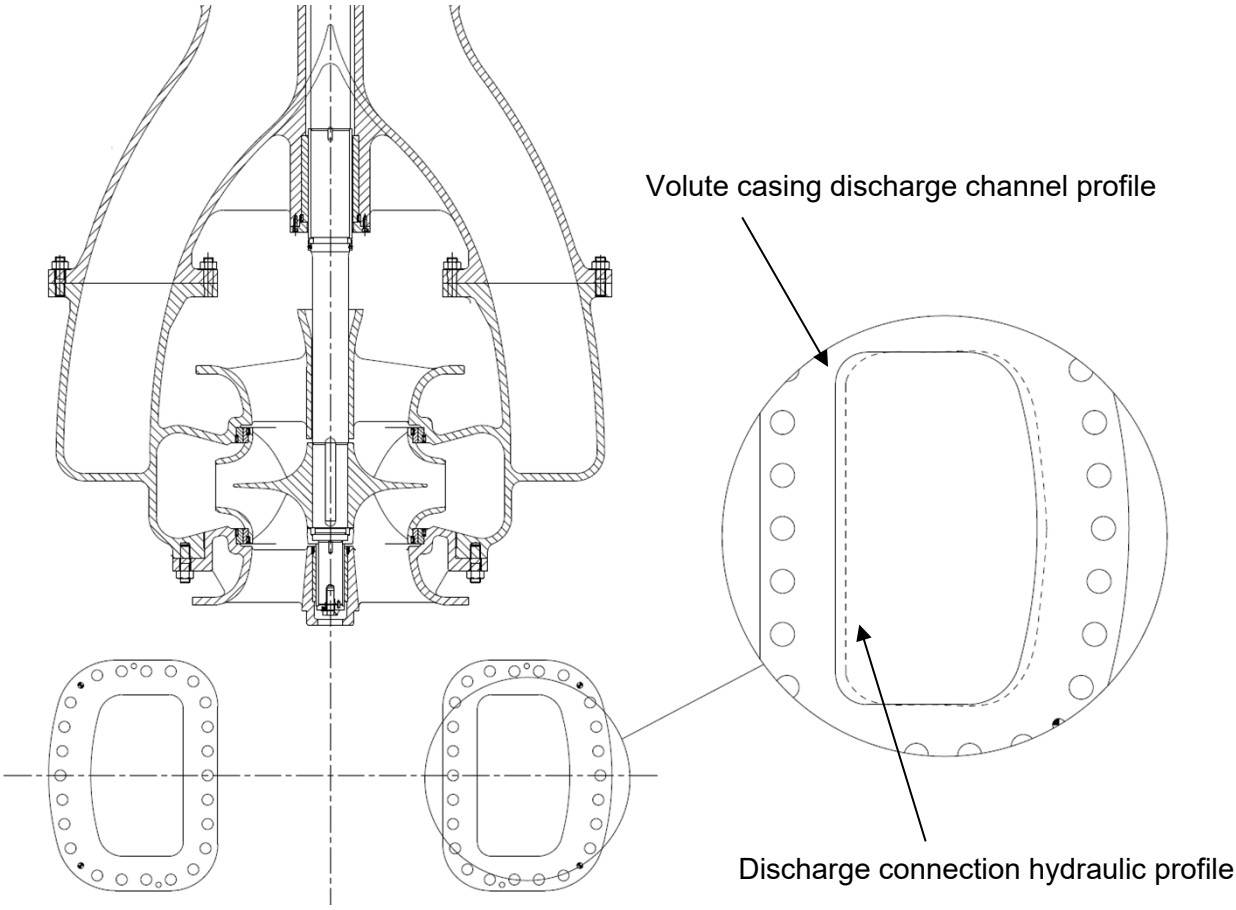

Figure 3: Example of hydraulic misalignment on VS2/VS7 discharge connection.

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Based on the pump type, the expected performance, the severity of the operating conditions, and the contractual tolerances applicable to the measured performance, it must be established whether discontinuities in the hydraulic channel require inspection and elimination or mitigation if necessary.

PT shall explicitly state in the TS, in accordance with the considerations outlined in the following Chapter 4, whether hydraulic continuity is to be verified and corrected if necessary.

4. Guidelines for hydraulic continuity verification

The hydraulic continuity of pumps should be verified and corrected when one of the following conditions is present:

- The pump design exists and has previously exhibited performance issues (e.g., efficiency losses, vibrations, etc.).
- The pump is classified as high energy according to **SOP-489** (ref. [1]).
- The required efficiency is high, near or at the top of the category, considering the pump type and size (see *Table 1*).
- The acceptance tolerances for performance do not permit any negative tolerance on efficiency: grade 1U or 1E according to ISO 9906 (ref. [3]) or HI 14.6 (ref. [4]).
- The acceptance tolerances for performance are according to API 610 (ref. [2]) or grade 1B according to ISO 9906 [3] or HI 14.6 [4]. Even when the contractual tolerance is not the most stringent factor, efficiency is known to be a critical issue for specific projects (e.g., efficiency increase projects).

For low energy, existing design BB1 pumps with no recorded performance issues, the verification may be limited to the discharge volute, omitting the suction chambers.

BB3 pumps are more critical because, due to the multistage configuration, discontinuities in the hydraulic channels at the joint plane produces cumulative losses and increased risk of flow separation in interstage channels, and because the typically smaller dimensions of the volute channels lead to increased sensitivity to geometric mismatches at the joint plane.

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Nq	Recommended efficiency limits for continuity check		
	BB1	BB3	VS2/VS7
10 - 20	> 80%	> 72%	> 78%
20 - 35	> 85%	> 76%	> 82%
35 - 50	> 87%	> 75%	> 80%
50 - 70	> 85%	> 70%	> 76%
>70	> 78%		> 75%

Table 1: Recommended efficiency limits for continuity check

5. Allowable limits

The maximum allowable discontinuity, measured on the joint plane, in any direction normal to the hydraulic profile (see *Figure 4*) is shown in Table 2, based on the pump size as determined by the minimum width of its volute channel at the joint plane.

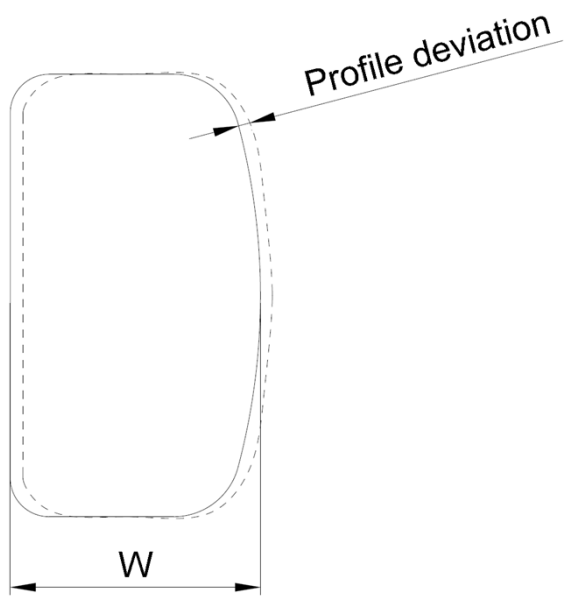

Figure 4: Deviations in the direction normal to the hydraulic profile.

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Volute channel width	Maximum allowable discontinuity	
	BB1/VS2/VS7	BB3
W ≤ 30 mm	0.8 mm	0.6 mm
30 < W ≤ 60 mm	1.0 mm	0.8 mm
60 < W ≤ 100 mm	1.2 mm	1.0 mm
100 < W ≤ 150 mm	1.5 mm	1.2 mm
W > 150 mm	2.0 mm	1.5 mm

Table 2: Maximum allowable discontinuity.

6. Measurement and correction of deviations

6.1. Machining of the hydraulic profile at the joint plane

If the hydraulic continuity check is specified in the TS, ME must assess whether it is feasible (from the technical or scheduling point of view) to perform the machining of the hydraulic profile to a depth based on the pump dimensions and the machining feasibility, in accordance with the hydraulic layout of the pump casing.

If machining is carried out, PA must verify that the operation has left no discontinuities; if discontinuities are present, all of them must be blended by grinding[1] until they fall within the tolerances specified in Table 2.

If machining is not feasible or not practical, the discontinuities must be measured and, if necessary, corrected as described in the following paragraph 6.2.

6.2. Verification and correction of continuity during assembly

The measurement of the deviations must be performed at assembly[1] as follows:

- Place a transparent tracing sheet over the cleaned mating surface of the lower casing. The position of the sheet must be fixed using the centering pins between the lower and upper casing as reference points (*Figure 5*).

[1] Detailed instructions on process, tools, safety precautions, etc. are not within the scope of this procedure.

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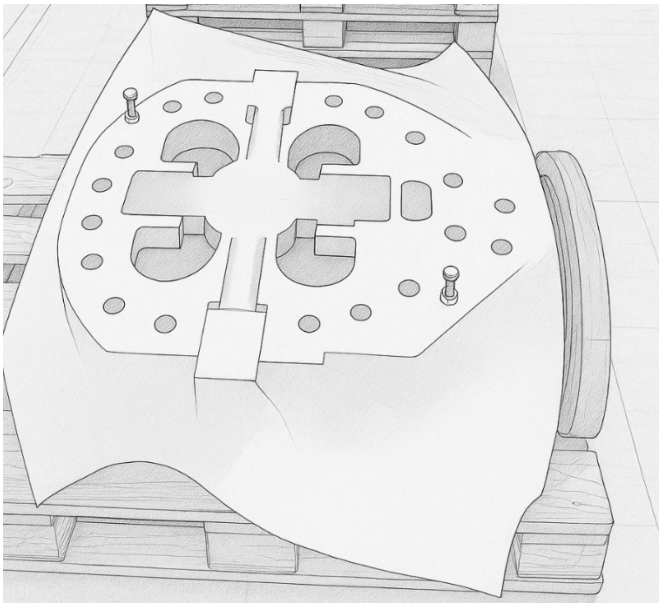

Figure 5: Positioning of the tracing sheet.

- Trace the outline of the hydraulic profile on the transparent tracing sheet using a marker (*Figure 6).*

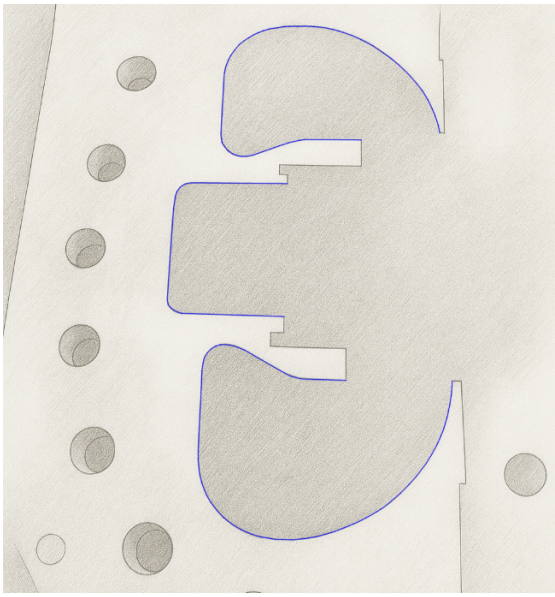

Figure 6: Tracing of the outline of the hydraulic profile.

- Repeat the operation by repositioning the sheet on the upper casing, aligning it with the centering pins.

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- At the end of the operation, the sheet will display the superimposed profiles of both casings (Figure 7).

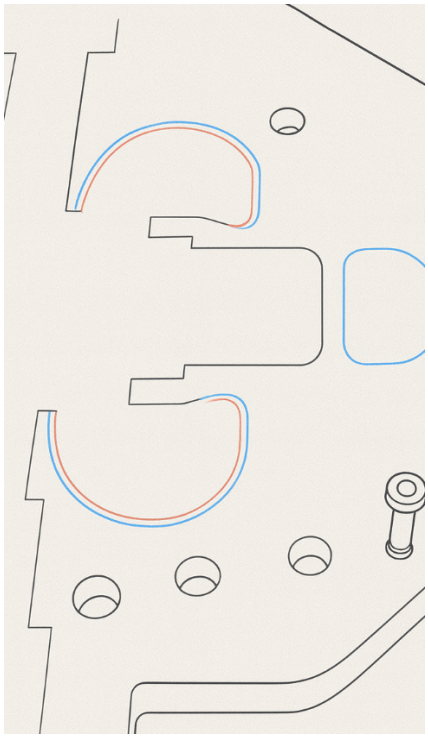

Figure 7: Tracing sheet with the superimposed profiles.

- Measure the deviations in the direction normal to the profile (ref. Chapter 5).

All measured deviations beyond the allowable limits (see *Table 2*) must be corrected as follows:

- Position the tracing sheet on the lower casing, using centering pins as reference, and mark with punch marks all areas of excess material to be removed (*Figure 8* and *Figure 9*).
- Repeat the marking on the upper casing.
- Remove all the excess material by grinding to reduce the deviation within the allowable tolerances.

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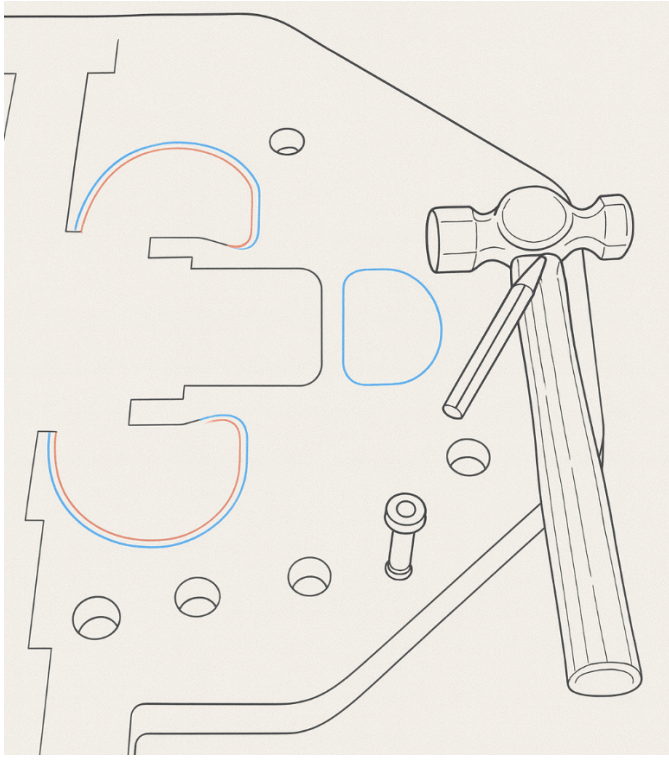

Figure 8: Repositioning of the tracing sheet in the lower casing.

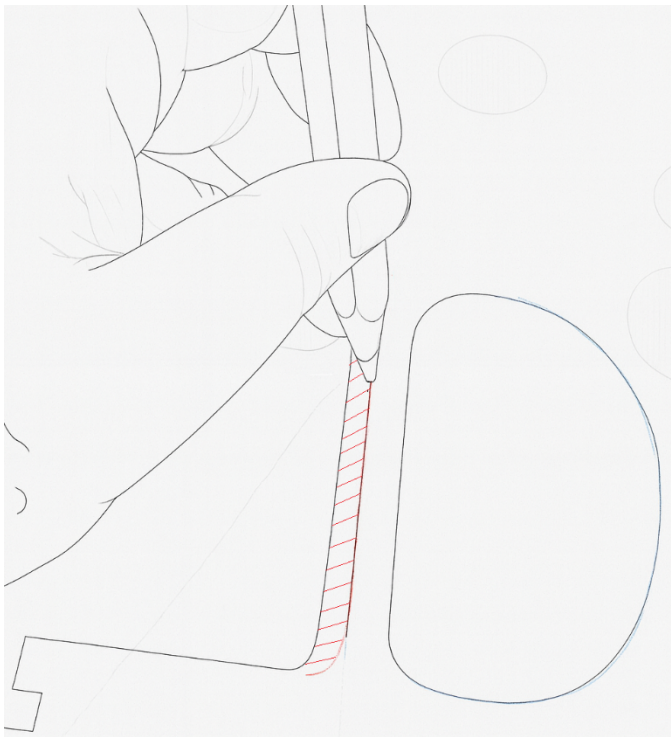

Figure 9: Marking of excess material to be removed.

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6.3. *Pumps with internal coating*

If the pump is designed for potable water service and the hydraulic passages are internally coated with a special paint such as Belzona, Chesterton, or an equivalent product, the hydraulic continuity check must be performed before applying the coating.

7. Reference documentation

[1] **SOP-489**, High-Medium-Low Energy pumps categorization.

[2] API Standard 610 12th Edition, Centrifugal Pumps for Petroleum, Petrochemical and Natural Gas Industries, American Petroleum Institute, 2021.

[3] EN ISO 9906:2012, Rotodynamic pumps - Hydraulic performance acceptance tests - Grades 1, 2 and 3, European Committee for Standardization, 2012.

[4] Hydraulic Institute, ANSI/HI 14.6-2022 - American National Standard for Rotodynamic Pumps for Hydraulic Performance Acceptance Tests, American National Standard Institute, 2022.

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